

AI Capability Building Workshop

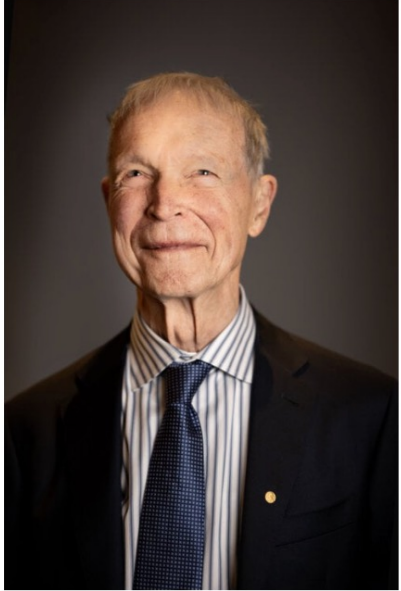
Bandara Dissanayake, PhD

December 10, 2024

Stockholm Concert Hall



Nobel Prize in Physics 2024



© Nobel Prize Outreach. Photo:
Nanaka Adachi

John J. Hopfield

Prize share: 1/2



© Nobel Prize Outreach. Photo:
Clément Morin

Geoffrey Hinton

Prize share: 1/2

The Nobel Prize in Physics 2024 was awarded jointly to John J. Hopfield and Geoffrey Hinton "for foundational discoveries and inventions that enable machine learning with artificial neural networks"

Nobel Prize in Chemistry 2024



© Nobel Prize Outreach. Photo:
Clément Morin

David Baker

Prize share: 1/2



© Nobel Prize Outreach. Photo:
Clément Morin

Demis Hassabis

Prize share: 1/4



© Nobel Prize Outreach. Photo:
Clément Morin

John Jumper

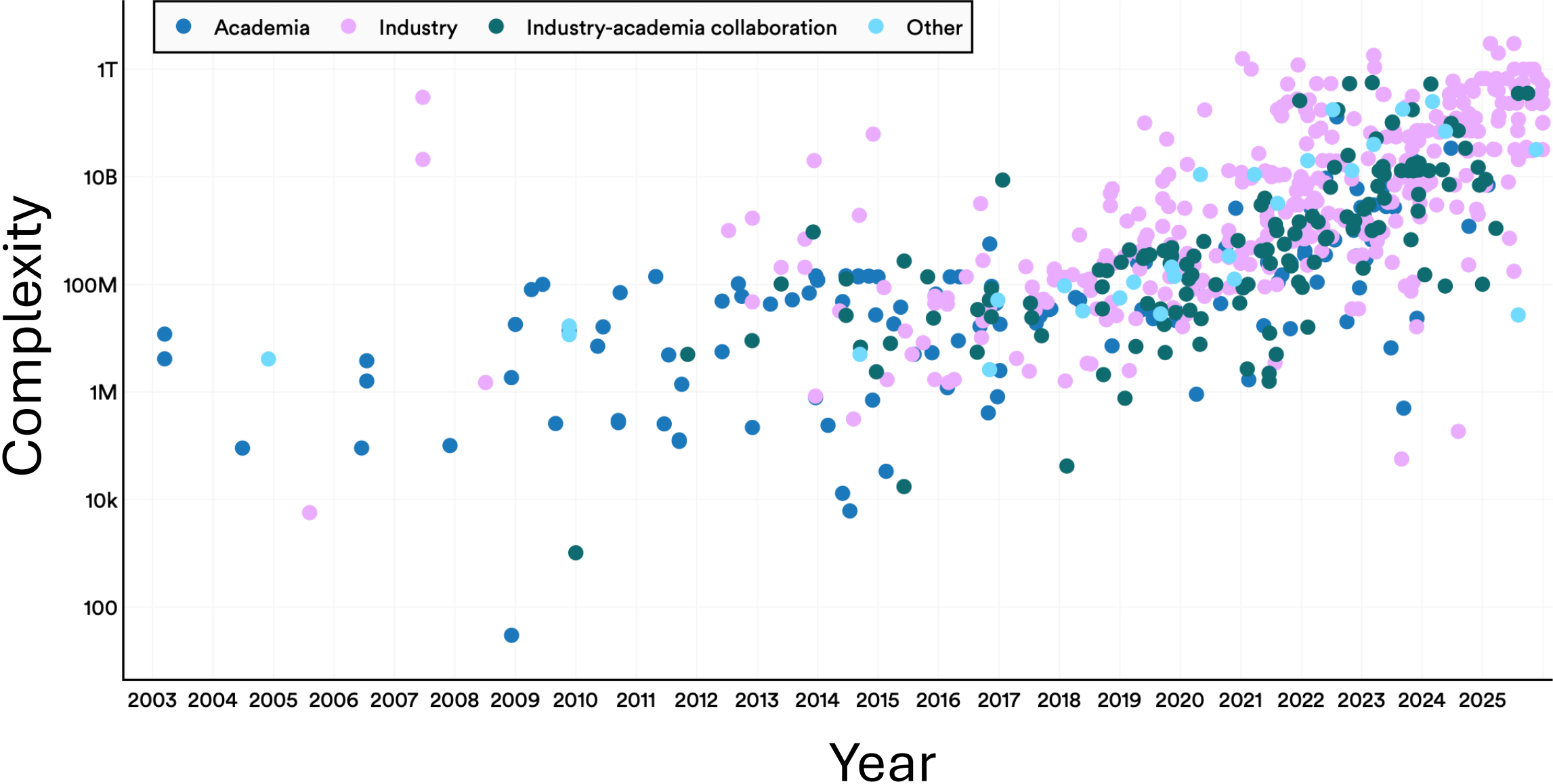
Prize share: 1/4

The Nobel Prize in Chemistry 2024 was divided, one half awarded to David Baker "for computational protein design", the other half jointly to Demis Hassabis and John Jumper "for protein structure prediction"

Is it all academics?

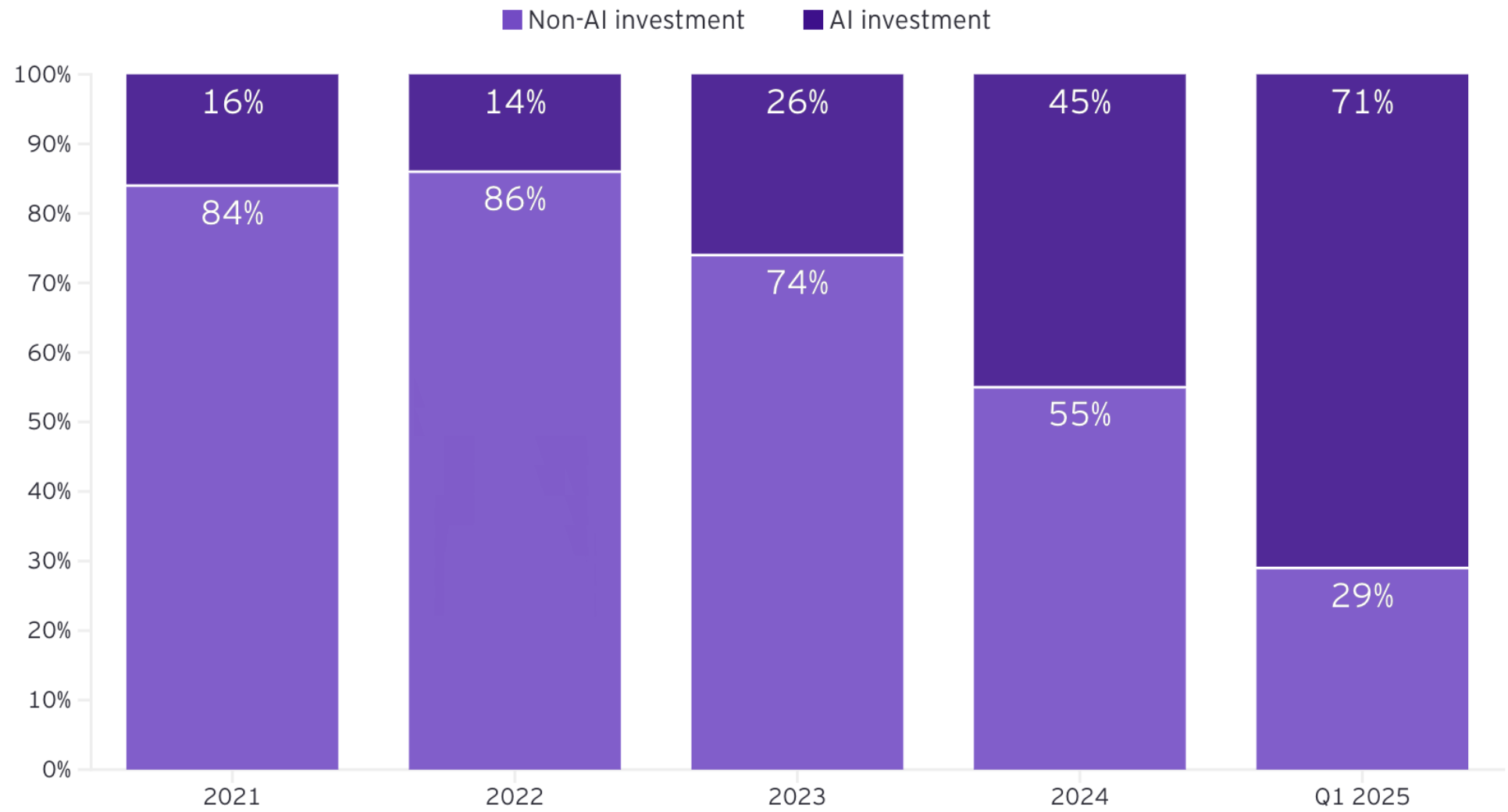
Number of parameters of notable AI models by sector, 2003–25

Source: Epoch AI, 2026 | Chart: 2026 AI Index report



Investment in AI

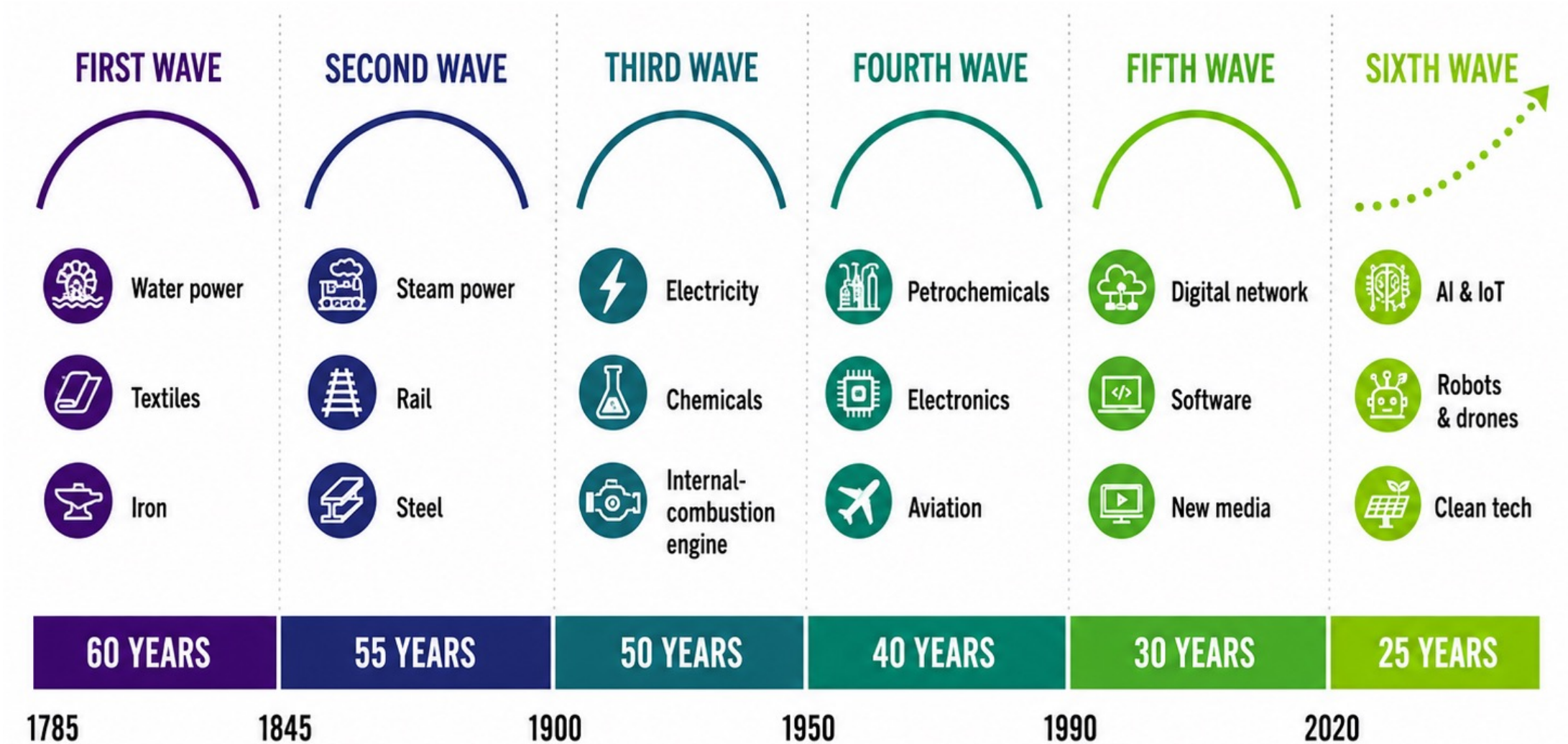
US VC-backed companies



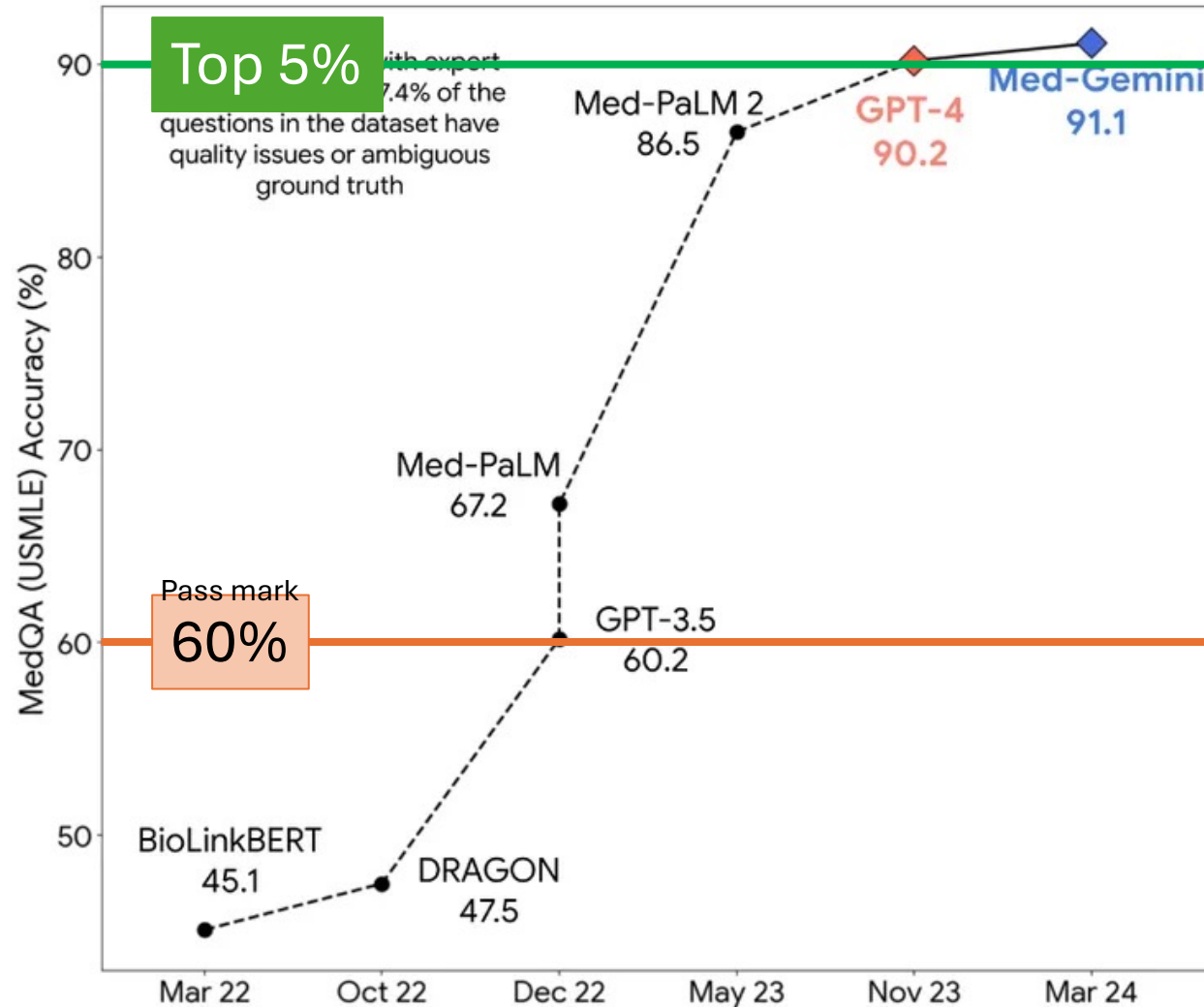
Source: EY analysis; Crunchbase. www.crunchbase.com, accessed April 6, 2025. Figures are for equity financings into VC-backed companies headquartered in the US. Cash investments came from VC firms, corporate investors, other private equity firms, individuals and other sources.

Why ???

Long Waves of Innovation



US Medical Licensing Examination



The Mother of Every Living Human

Human mitochondria hold a second genome, which is inherited from the mother



Mitochondrial Eve is the woman all living humans inherited their mitochondrial DNA from

She lived 200,000 years ago, and her mitochondrial lineage is called L0







Sapiens Dominated

Problem
Solving

Learning

Reasoning

Human Intelligence



If Nayana plucks 1 *sal* flower, how many left?



In the image provided, there are **three** open Sal flowers (Cannonball tree flowers) visible on the hanging stalk that the girl (Nayana) is reaching for.

If Nayana plucks one flower, there would be **two** flowers left on that specific cluster.

Reasoning:

- **Observation:** Looking closely at the stalk hanging down, you can see three distinct, fully bloomed pink and yellow flowers.
- **Action:** Plucking is a simple subtraction:
 $3 \text{ total flowers} - 1 \text{ plucked} = 2 \text{ remaining}$.
- **Context:** While there are many small buds further up the stalk, only three have fully opened into the characteristic Sal flower shape shown in the photo.



None of previous technologies
had this much of intelligence

Real Estate and Rental and Leasing



- Concierges
- Real Estate Sales Agents
- Real Estate Brokers
- Counter and Rental Clerks
- Property, Real Estate, & Community Association Managers

Government



- Recreation Workers
- Compliance Officers
- First-Line Supervisors of Police and Detectives
- Administrative Services Managers
- Child, Family, and School Social Workers

Manufacturing



- Mechanical Engineers
- Industrial Engineers
- Buyers & Purchasing Agents
- Shipping, Receiving, & Inventory Clerks
- First-Line Supervisors of Production and Operating Workers

Professional, Scientific, and Technical Services



- Software Developers
- Lawyers
- Accountants & Auditors
- Computer & Information Systems Managers
- Project Management Specialists

Health Care and Social Assistance



- Registered Nurses
- Nurse Practitioners
- Medical & Health Services Managers
- First-Line Supervisors of Office & Administrative Support Workers
- Medical Secretaries & Administrative Assistants

Finance and Insurance



- Customer Service Representatives
- Financial & Investment Analysts
- Financial Managers
- Personal Financial Advisors
- Securities, Commodities & Financial Services Sales Agents

Retail Trade



- Pharmacists
- General and Operations Managers
- Private Detectives & Investigators
- First-Line Supervisors of Retail Sales Workers

Wholesale Trade



- Sales Managers
- Order Clerks
- Sales Representatives, Wholesale & Manufacturing, Technical & Scientific Products
- Sales Representatives, Wholesale & Manufacturing, Except Technical & Scientific Products
- First-Line Supervisors of Non-Retail Sales Workers

Information



- Producers & Directors
- Film & Video Editors
- Editors
- News Analysts, Reporters, & Journalists
- Audio and Video Technicians

How well can it perform real professional work compared with humans?

GDPval-AA v2 Leaderboard

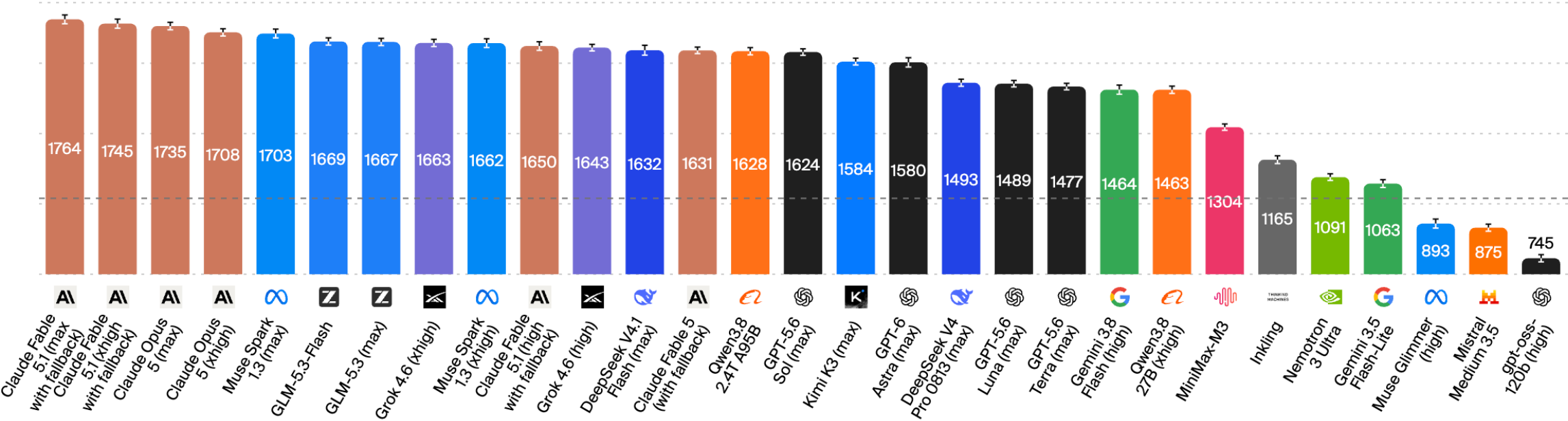
Elo rating for performance on real-world work tasks · Anchored to a human baseline of 1,000 · Higher is better

[↶](#)[🔍](#)[📊](#)

29 of 242 models [⬆](#)[🔍](#)[⚙️](#)

----- Human Baseline (1,000)

Artificial Analysis

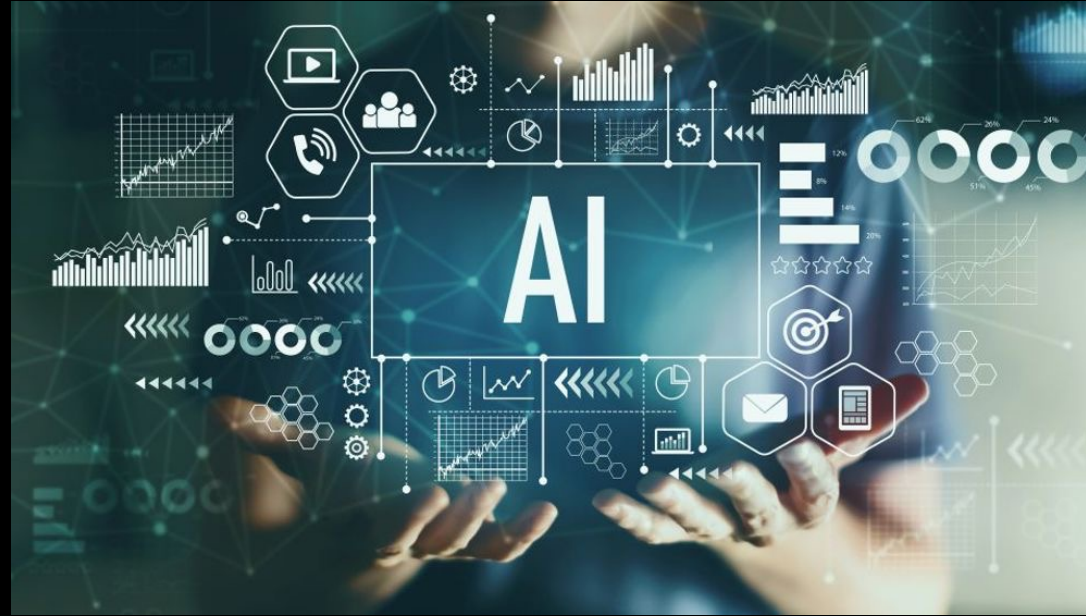


Benchmark: GDPval-AA v2 Elo

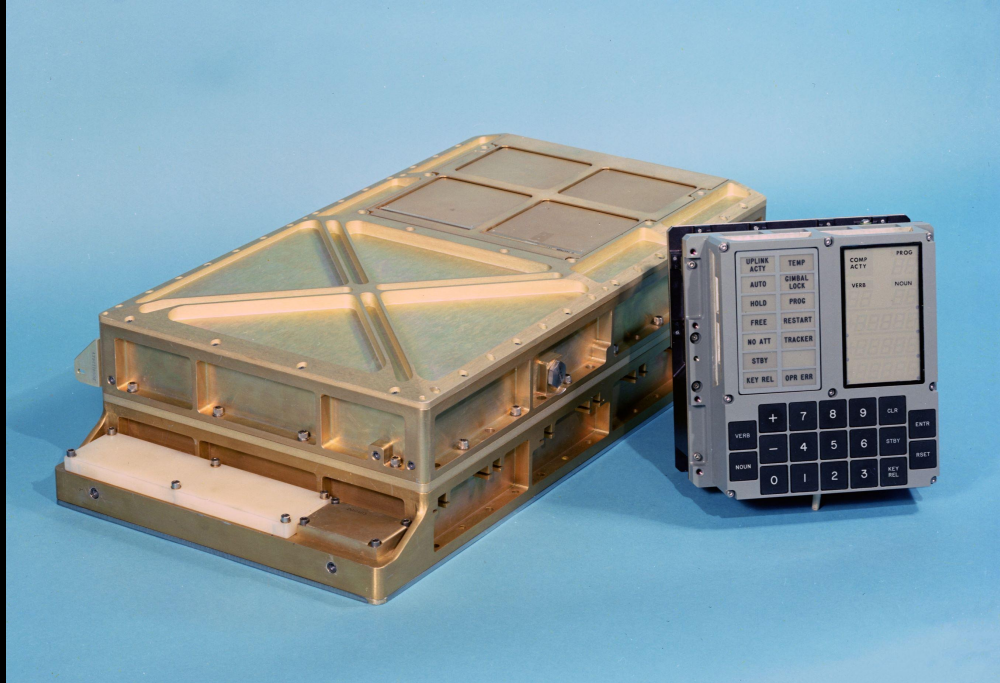
**AI is turning
into something
totally new**



Mustafa Suleyman, Microsoft AI CEO



We can't afford to ignore AI



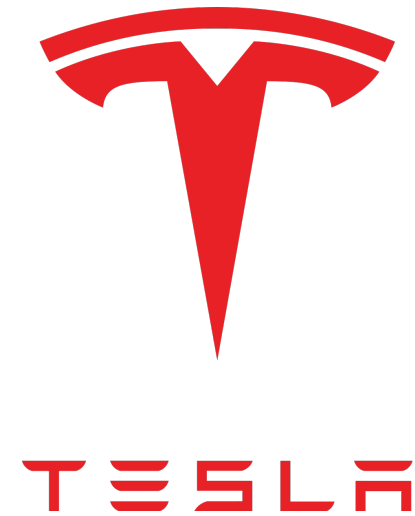
Apollo guidance computer 1960s





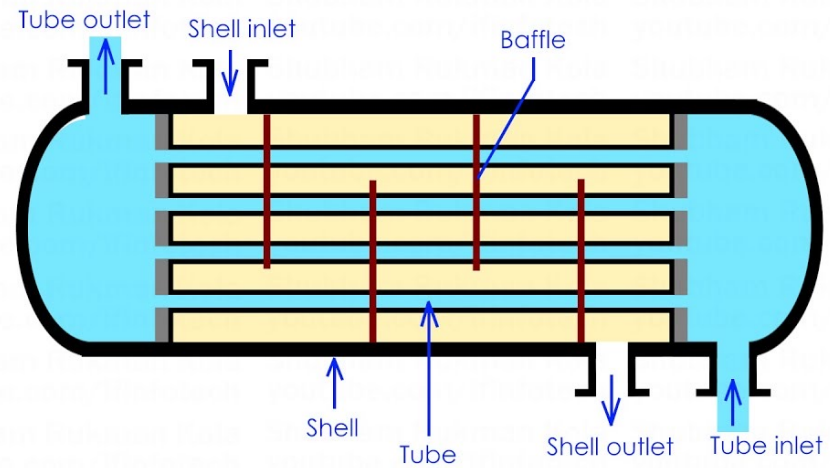
Just imagine the potential

If you don't do it, someone else will



*Tesla didn't build a better engine
They made the engine irrelevant*

Heat Exchanger





10x smaller, 50% more effective

How should we leverage
this to solve critical
business problems?

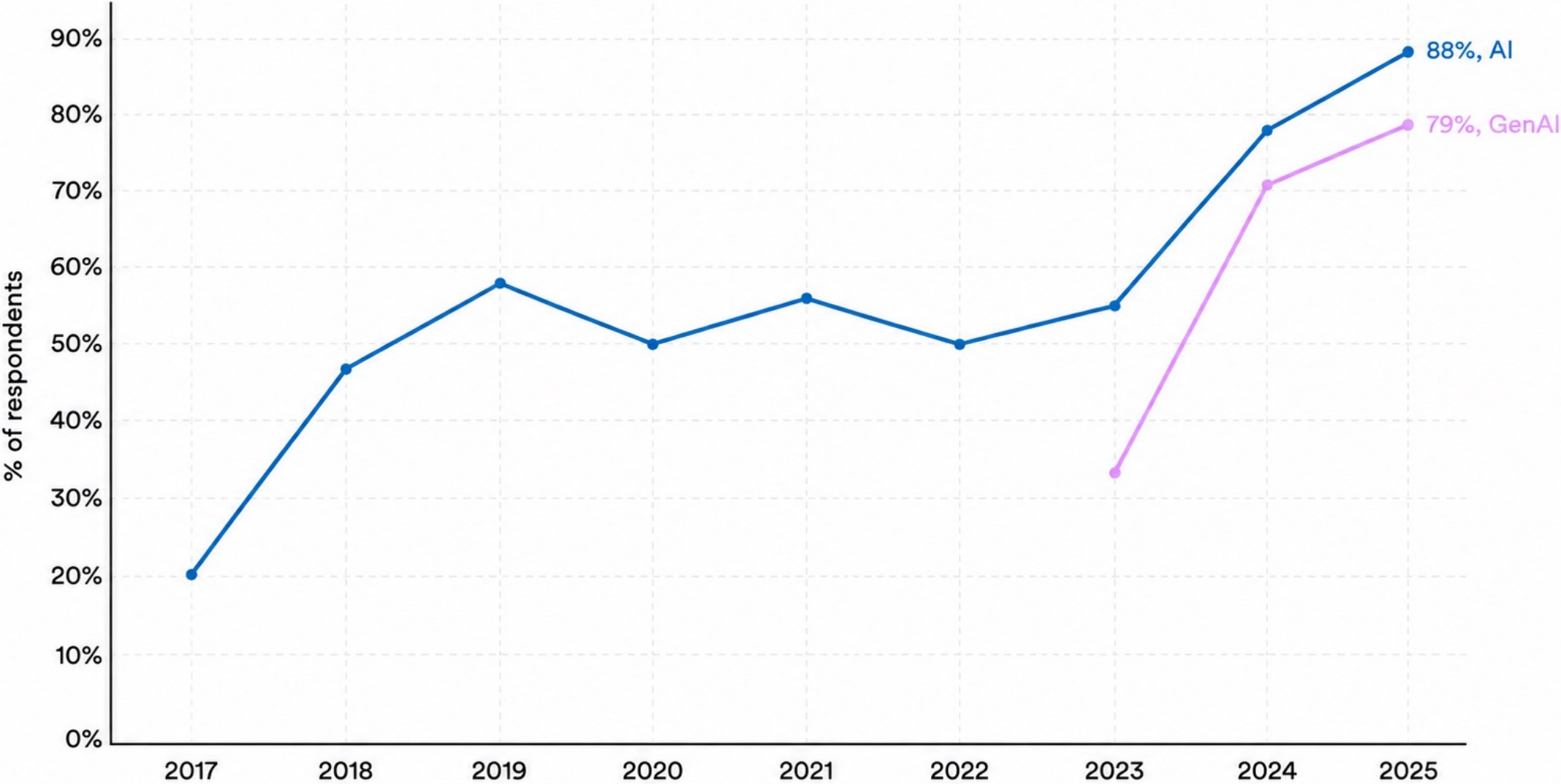
Chapter 2: Strategy

Let's give Copilot/
ChatGPT/Gemini access
to everyone



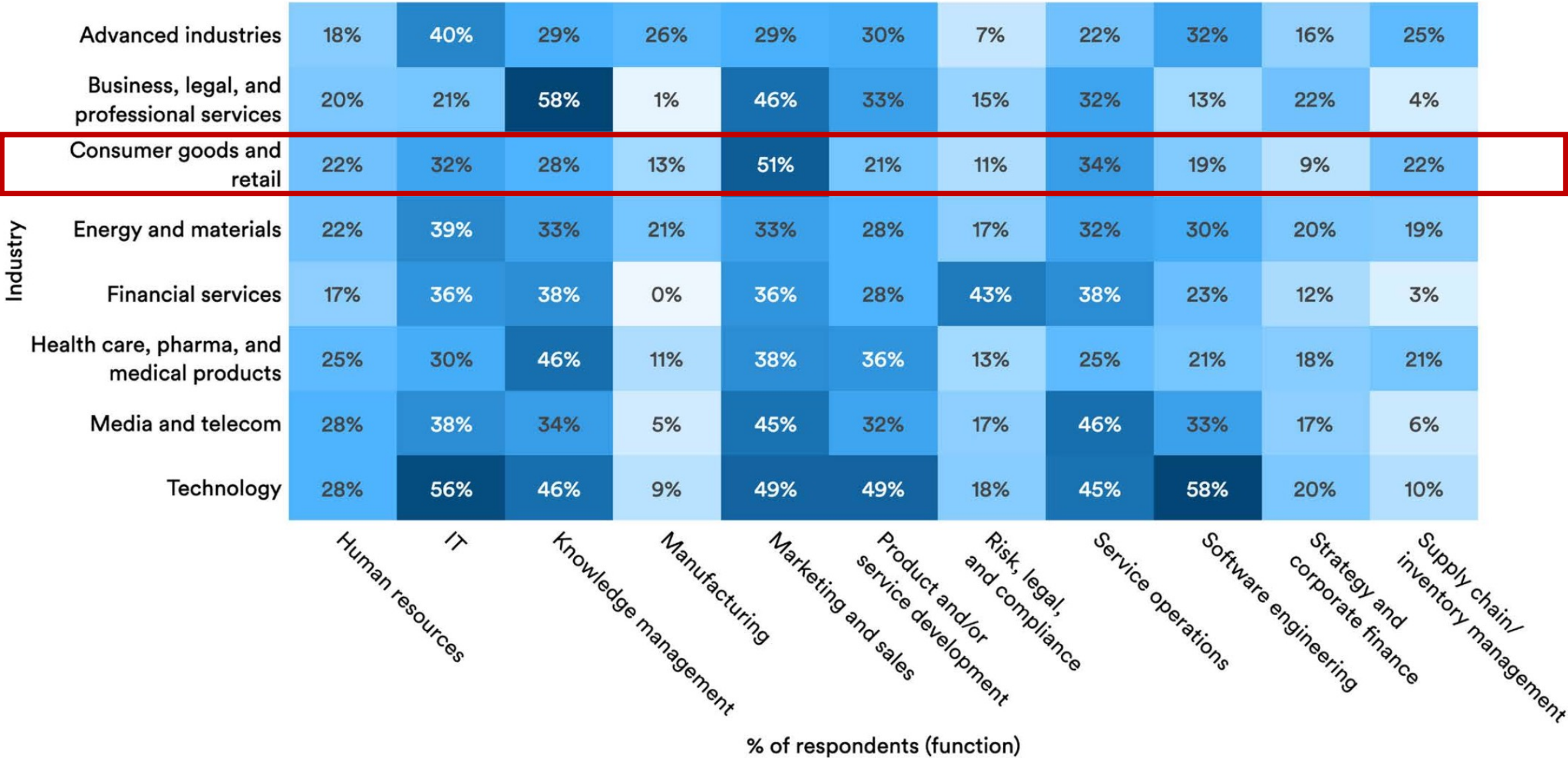
Organizations using AI in at least one function (2017–25)

Source: McKinsey & Company Survey, 2025 | Chart: 2026 AI Index report



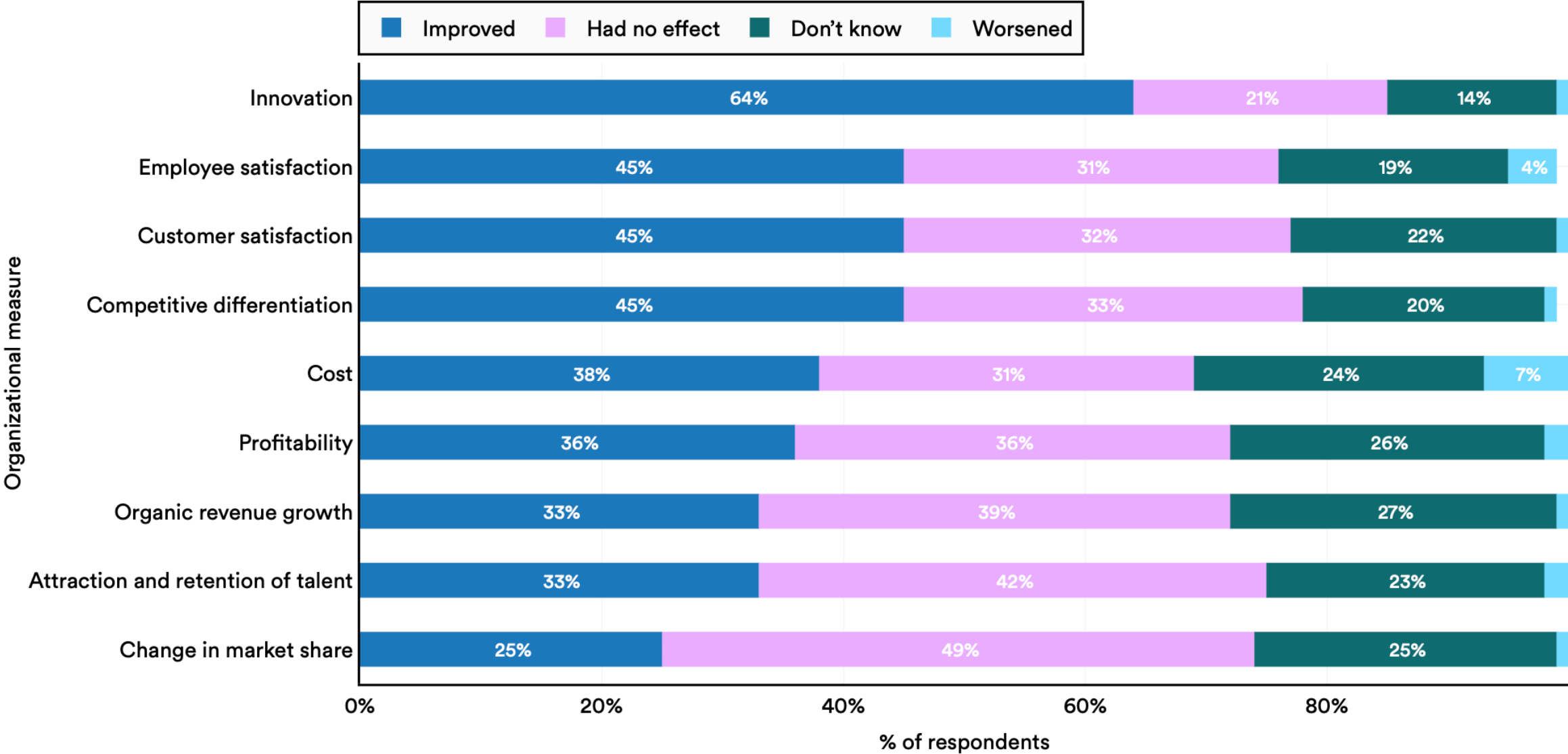
AI use by industry and function, 2025

Source: McKinsey & Company Survey, 2025 | Chart: 2026 AI Index report

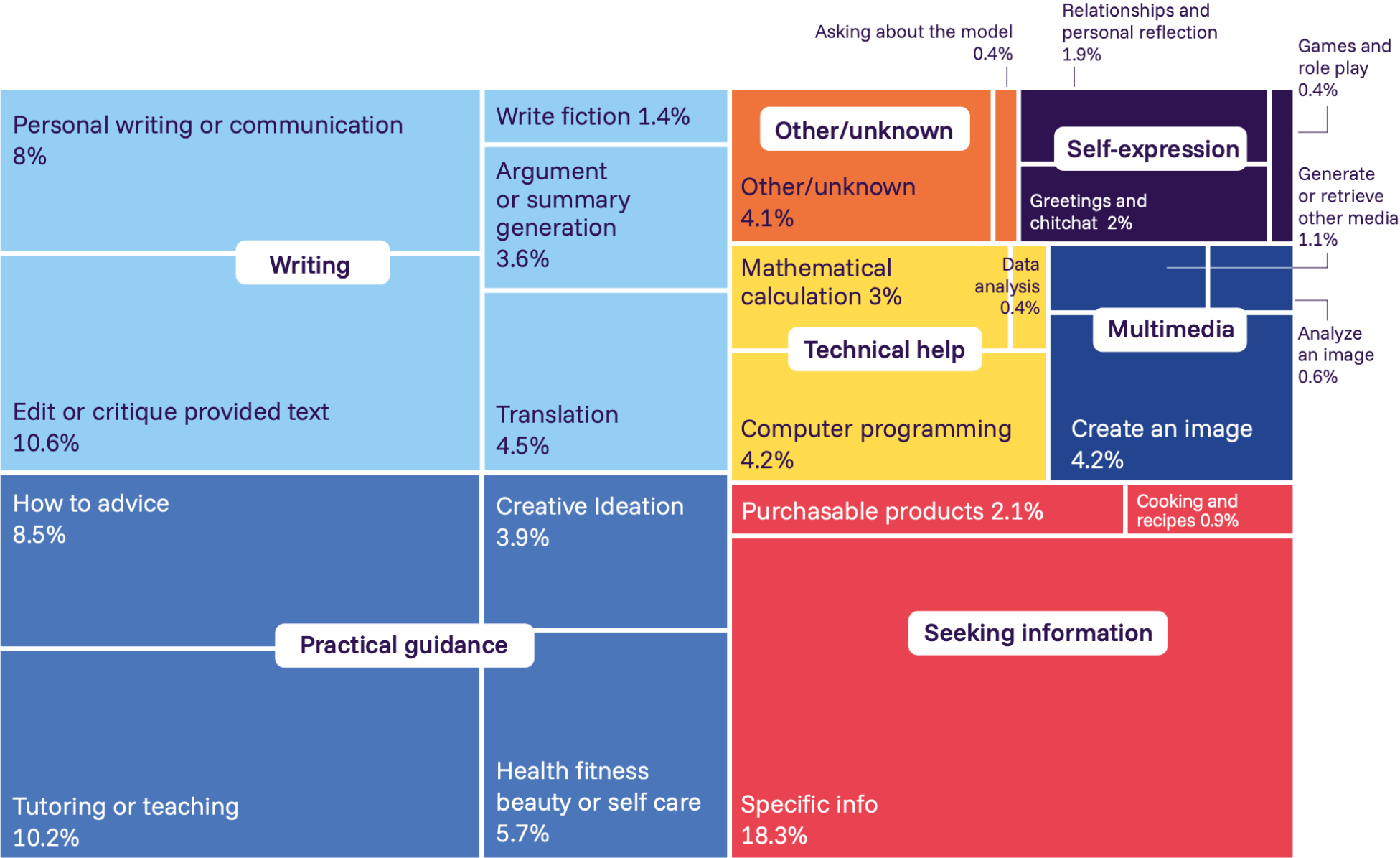


AI impact on organizational measures over the past year, 2025

Source: McKinsey & Company Survey, 2025 | Chart: 2026 AI Index report



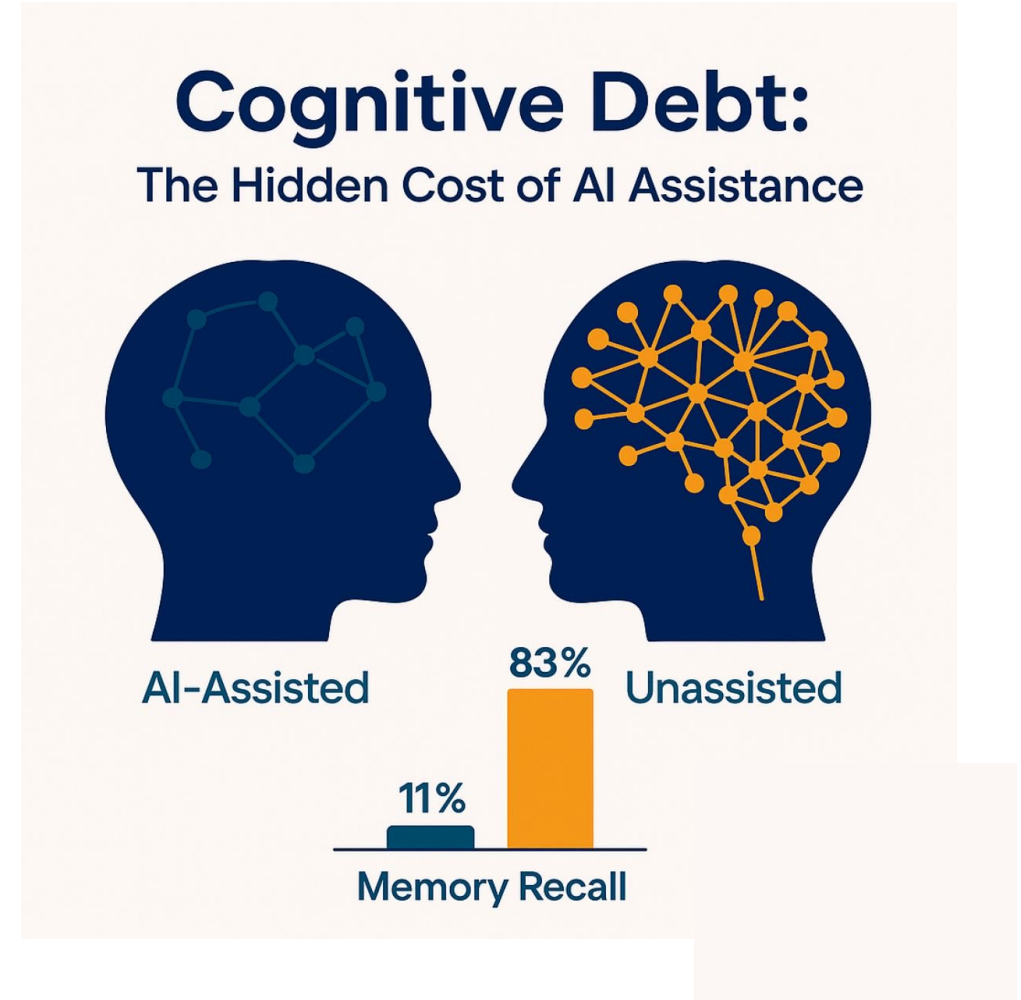
The most common uses of OpenAI’s ChatGPT are seeking information, writing, and practical guidance



Do we really know how
to leverage it?

- Lower brain activity
- Weaker brain connectivity
- Lower level of engagement

- Worse info recall
- Poor quality work
- Residual negative effect



Critical Thinking Erosion

Variable Pair	Correlation (r)
AI Tool Use ↔ Cognitive Offloading	+0.72
AI Tool Use ↔ Critical Thinking	−0.68
Cognitive Offloading ↔ Critical Thinking	−0.75

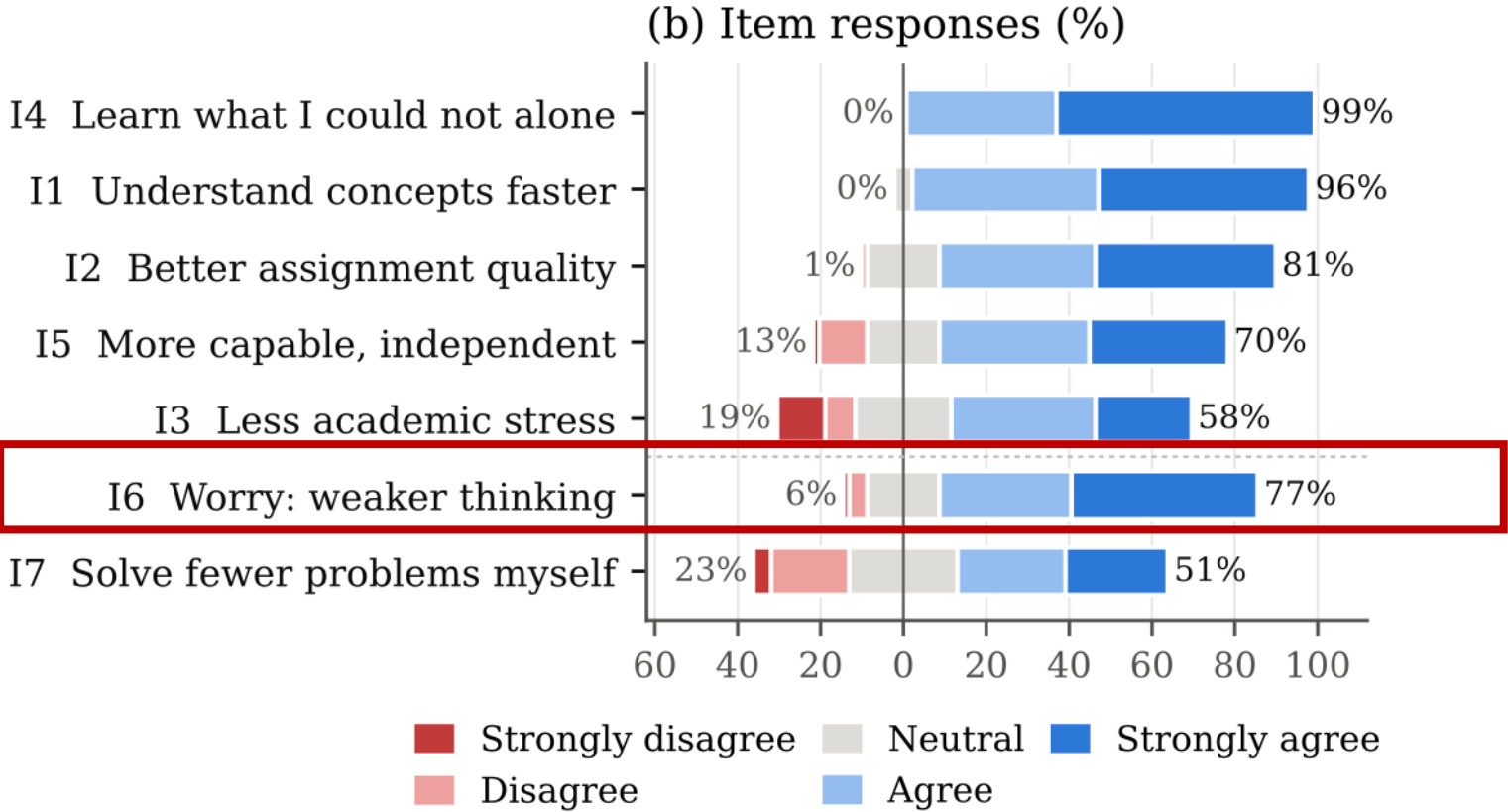
Article

AI Tools in Society: Impacts on Cognitive Offloading and the Future of Critical Thinking

Michael Gerlich 

Center for Strategic Corporate Foresight and Sustainability, SBS Swiss Business School,
8302 Kloten-Zurich, Switzerland; michael.gerlich@cantab.net

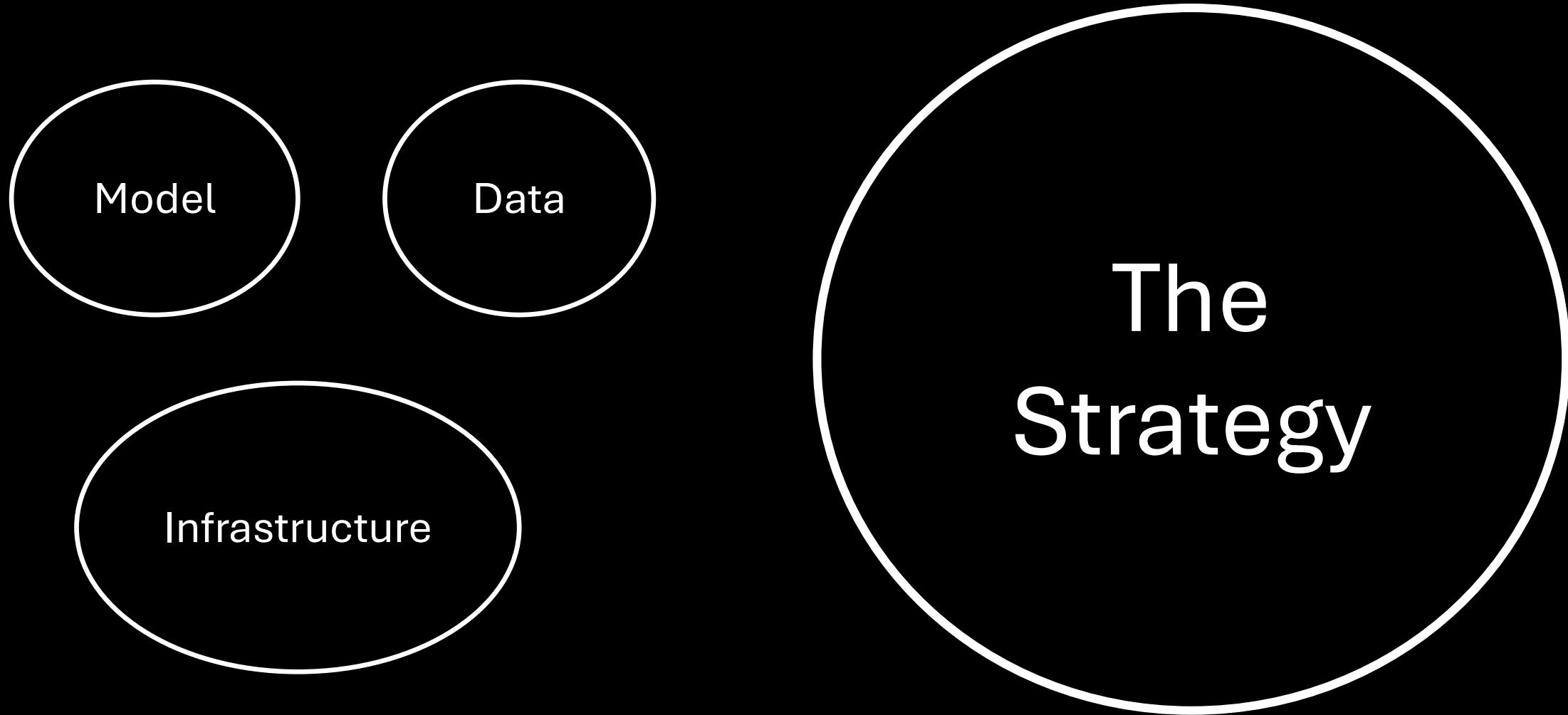
Self-Regulation, Problematic Social Media Use and Generative-AI use among Engineering Undergraduates



Business success
with eroding critical
thinking?

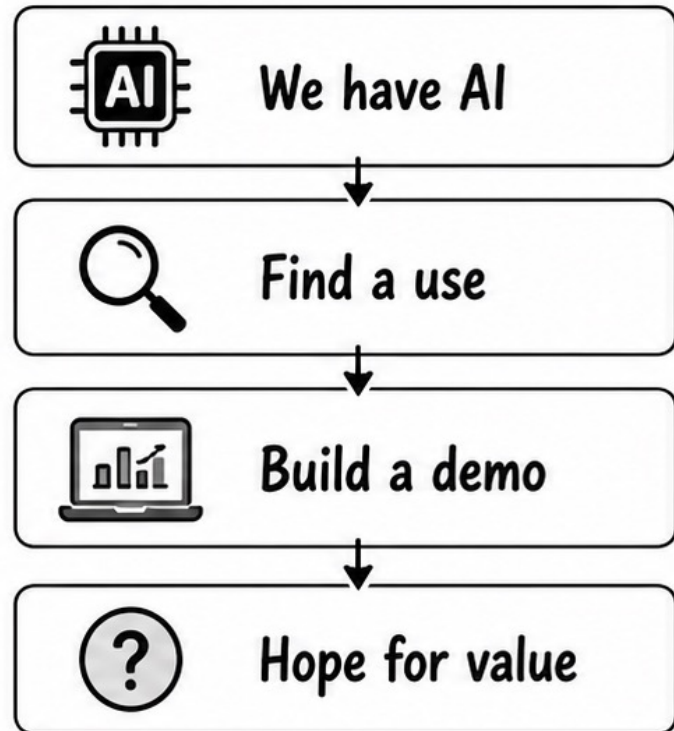


Integrating AI into Biz – Essential Components



TOOL-FIRST

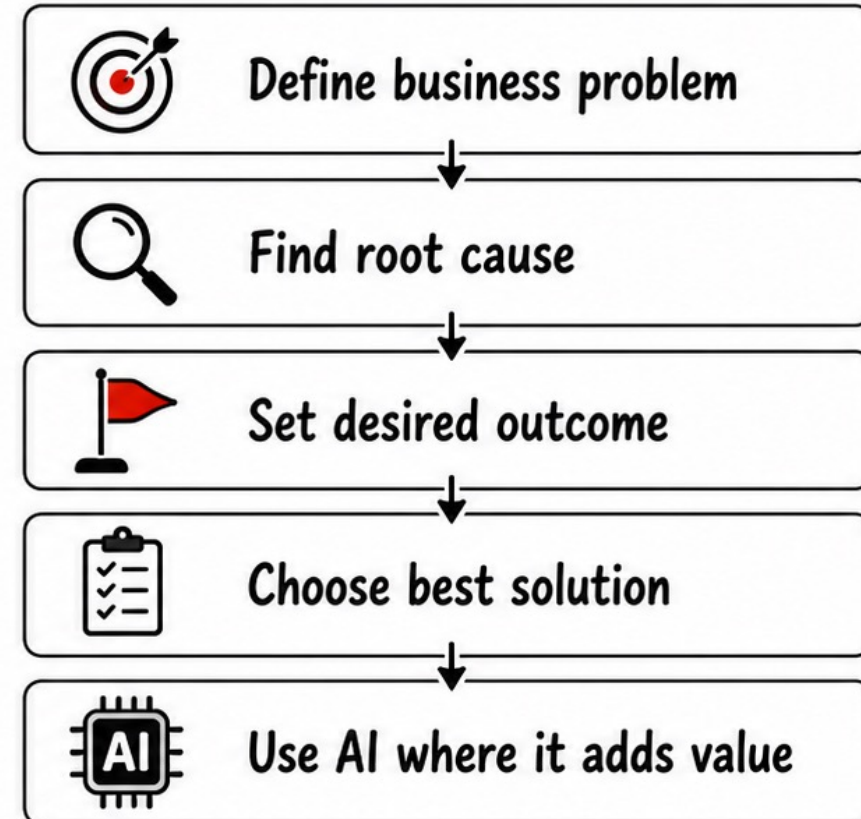
Start with the tool



⚠ Often leads to unclear value

BUSINESS-PROBLEM-FIRST

Start with the problem



✓ Focused on cost, quality, output, and decisions

THE MISCONCEPTION:

BUSINESS USE OF AI IS **JUST...**



Write
emails



Draft
documents



Make
presentations



Summarize
notes



AI CAN ALSO HELP WITH...



Personalized
training



Compare products /
suppliers



Extract insights
from reports



Analyze data &
find insights

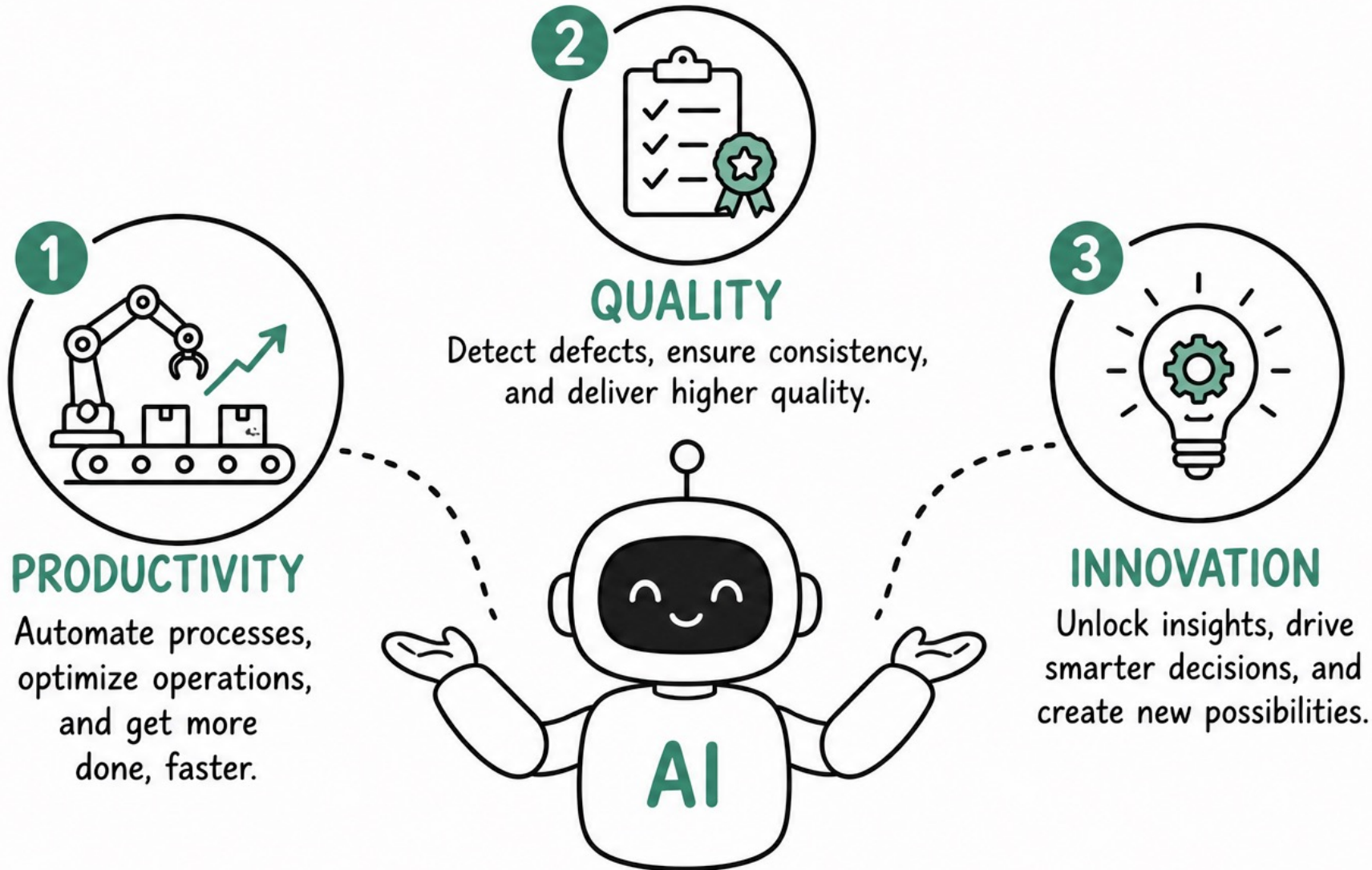


Forecast demand,
inventory &
cash flow



Optimize
workflows &
resources

AI APPLICATIONS IN MANUFACTURING



Chapter 3: The Definitions

Sinhala – detected

↔

English

×

යමු සෙල්ලකතරගම්,
ඔබ එන්න යනවා නම්,
කඳු කුමාරයා වල්ලිට
පෙම් බැන්ද කතරගම්
yamu sellakataragam, oba enna
yanavā nam, kaṇḍa kumārayā valliṭa
pem bænda kataragam
Did you mean: යමු සෙල්ලකතරගම්, ඔබ යන්...

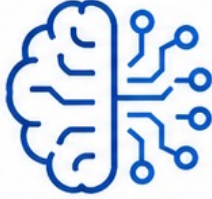
Let us go to
Kataragama—if you
are coming along—to
the Kataragama
where Lord Skanda
fell in love with Valli.

Is this an AI system?

Artificial Intelligence (AI)

Artificial intelligence (AI) is the capability of computational systems to perform tasks that typically require human intelligence, including **learning, reasoning, problem-solving, perception, language understanding, and decision-making**

AI – Different Branches



Artificial Intelligence (AI)

Computer systems that can perform tasks that normally require human intelligence.



Machine Learning (ML)

A subset of AI that enables computers to learn from data without being explicitly programmed.



Deep Learning (DL)

A subset of ML that uses artificial neural networks to learn complex patterns.



Generative AI (GenAI)

A type of deep learning that can create new content such as text, images, audio, or code.



Data

Models

**The
Economist**

MAY 6TH-12TH 2017

Crunch time in France

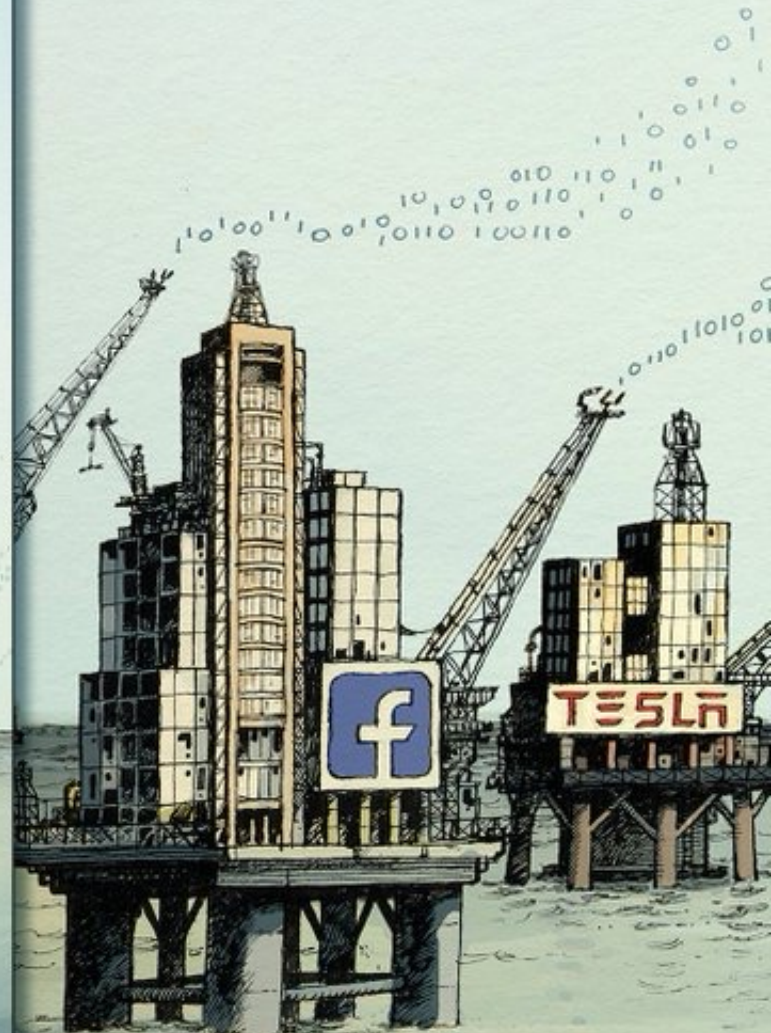
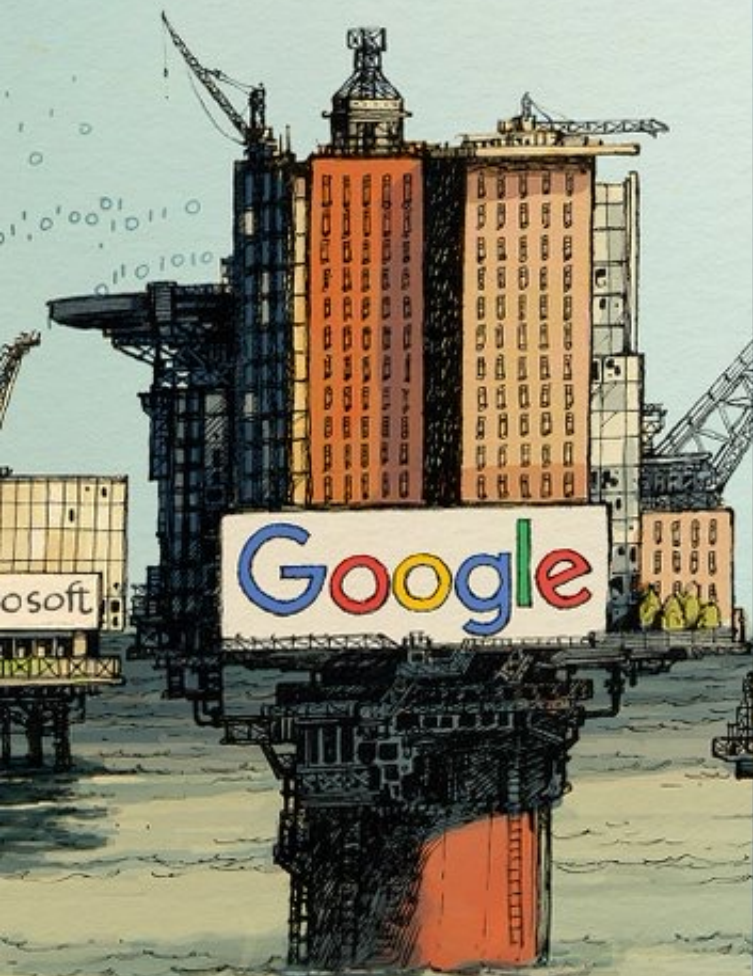
Ten years on: banking after the crisis

South Korea's unfinished revolution

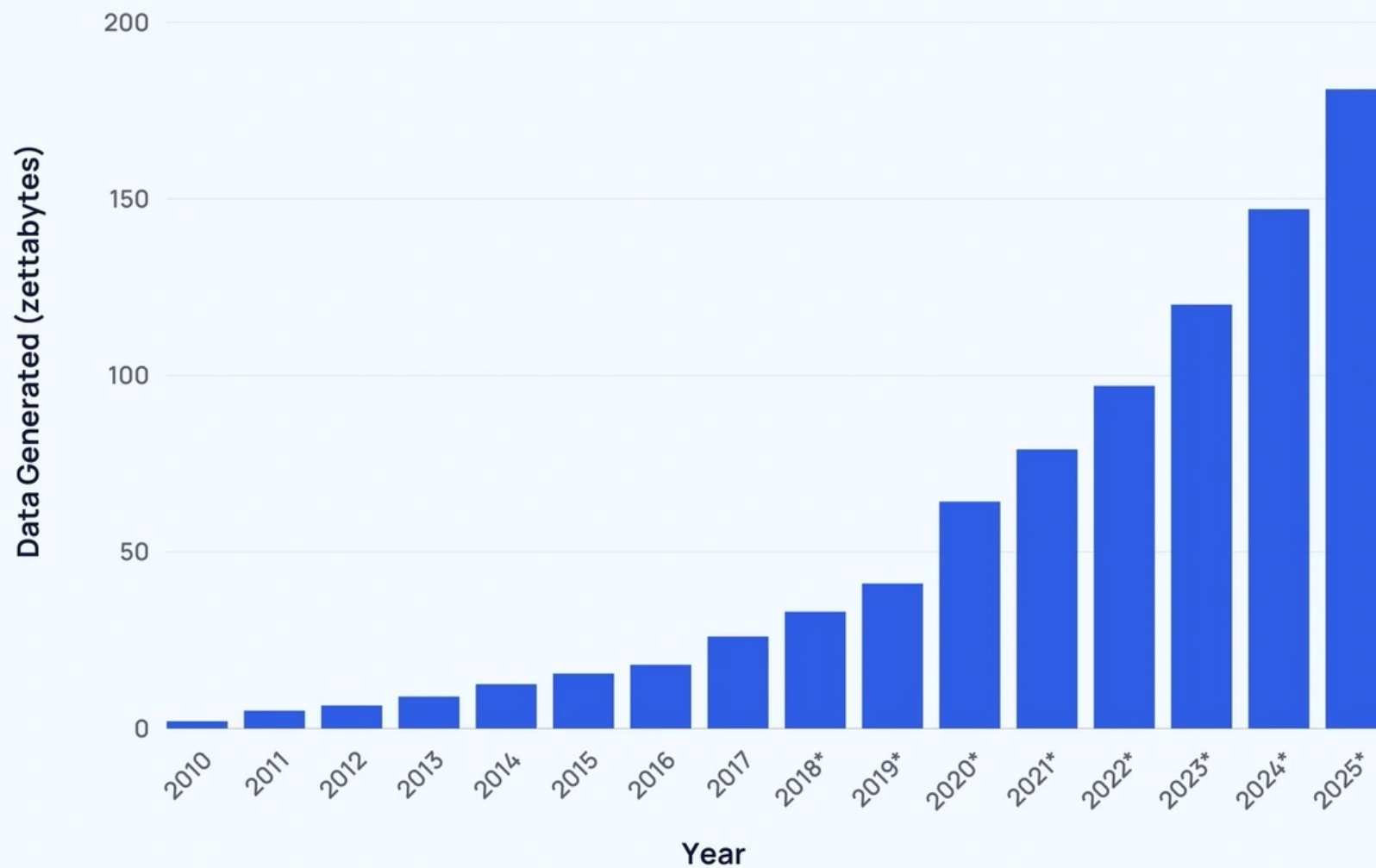
Biology, but without the cells

The world's most valuable resource

**Data and the new rules
of competition**



Global Data Generated Annually

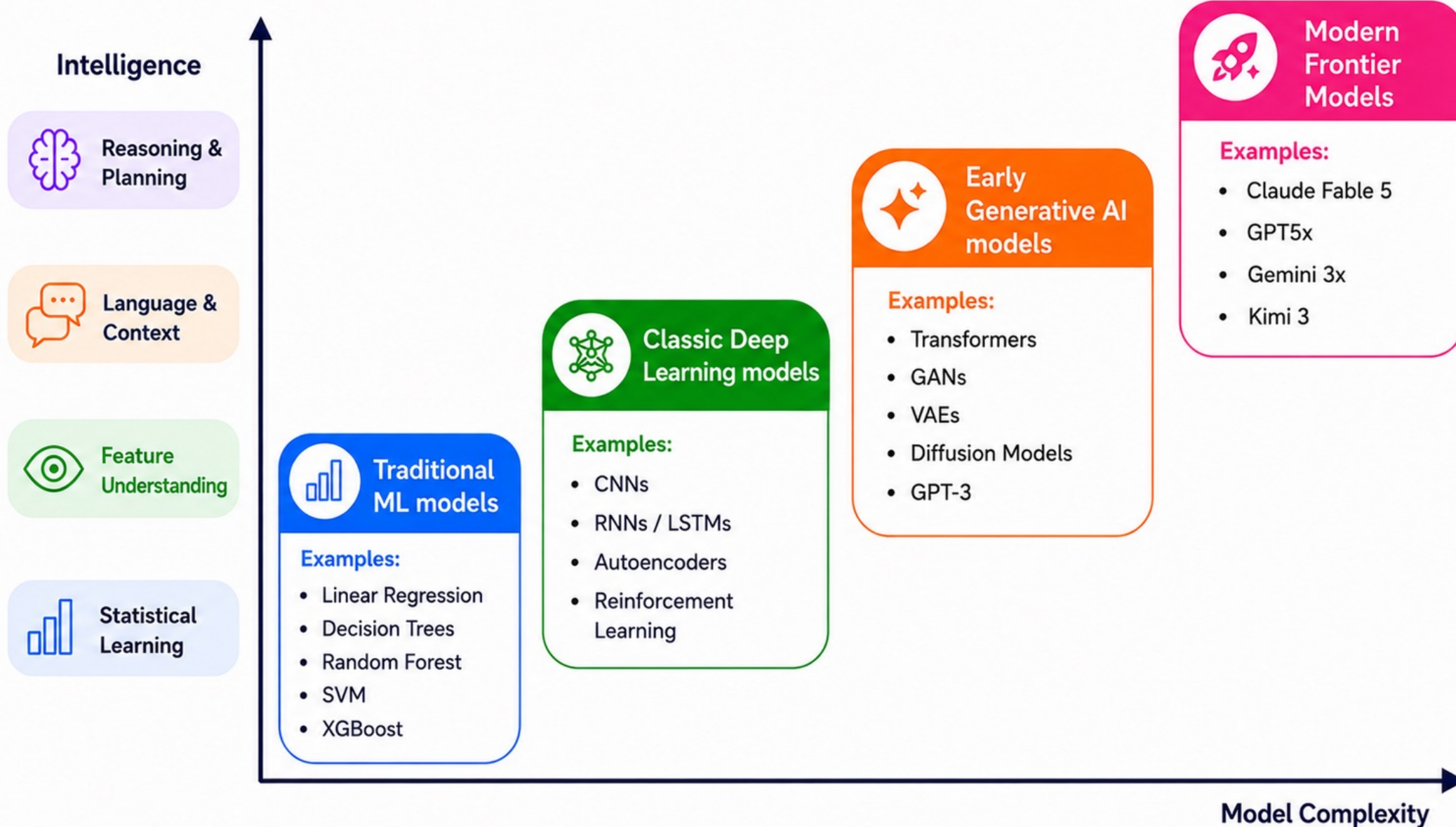


AI is data driven

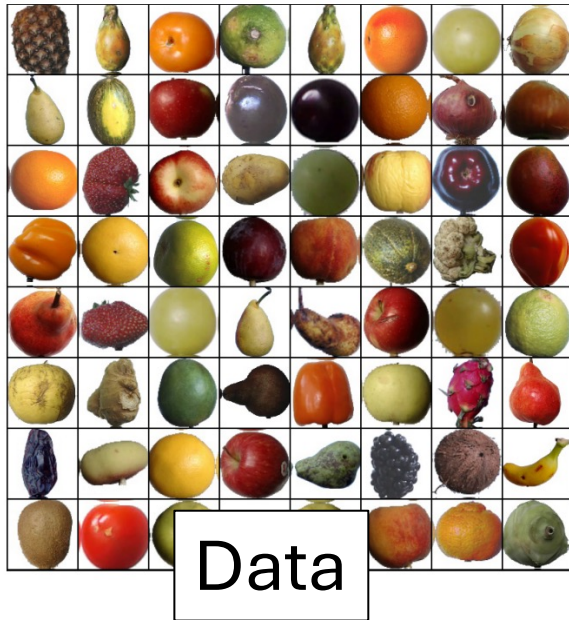
- Do you have data?
- If not, how do you collect data?
- How much data you have?
- What is the quality of the data?

Remember: **Data is not cheap**

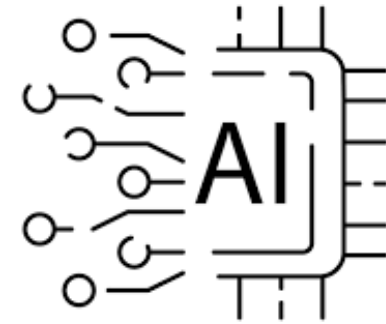
The Evolution of Models



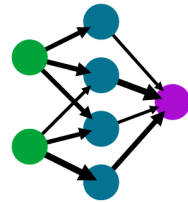
Training



Training

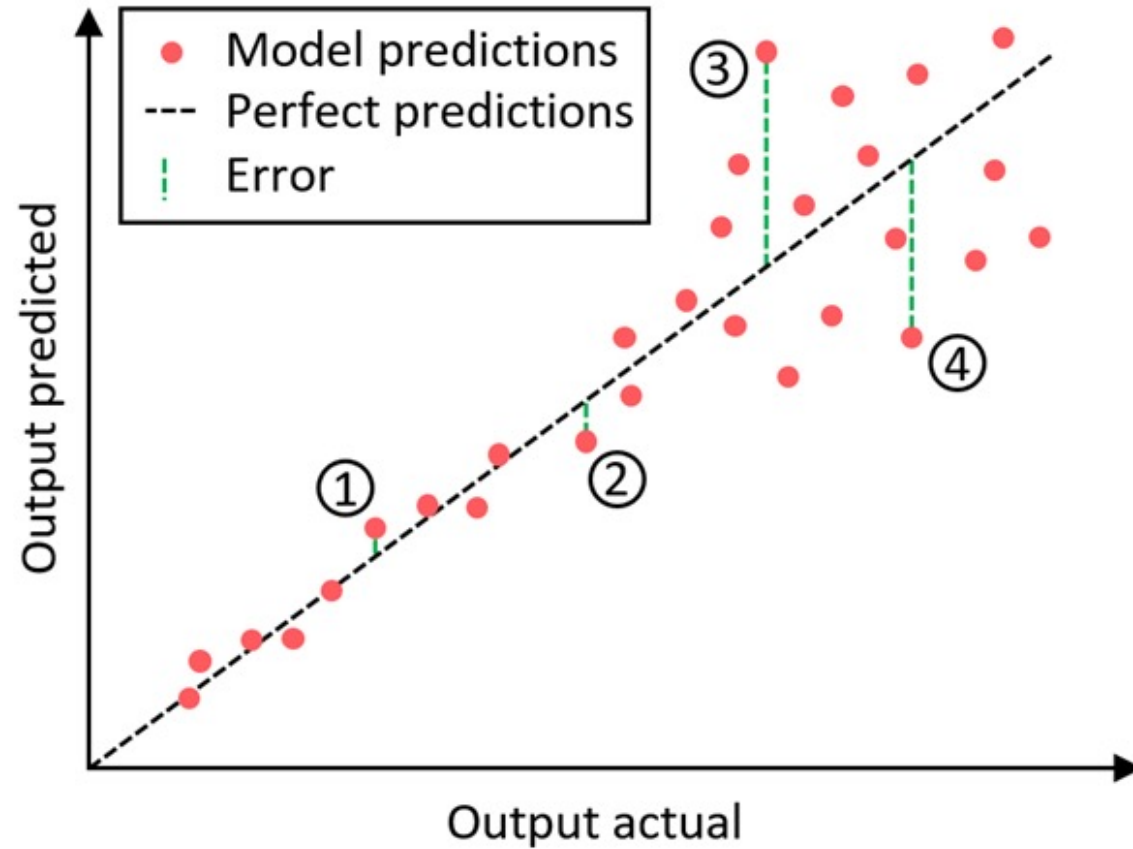


Trained AI Algorithm



Training: Teaching an AI model with data

Testing/Validation



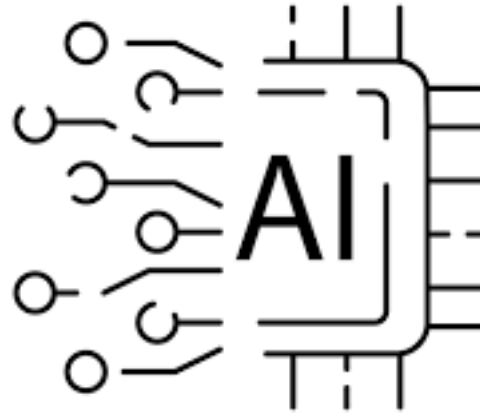
Higher the error, poor the model is

Validation: Ensure the model is accurate for the task

Inference



New data



Trained AI Algorithm

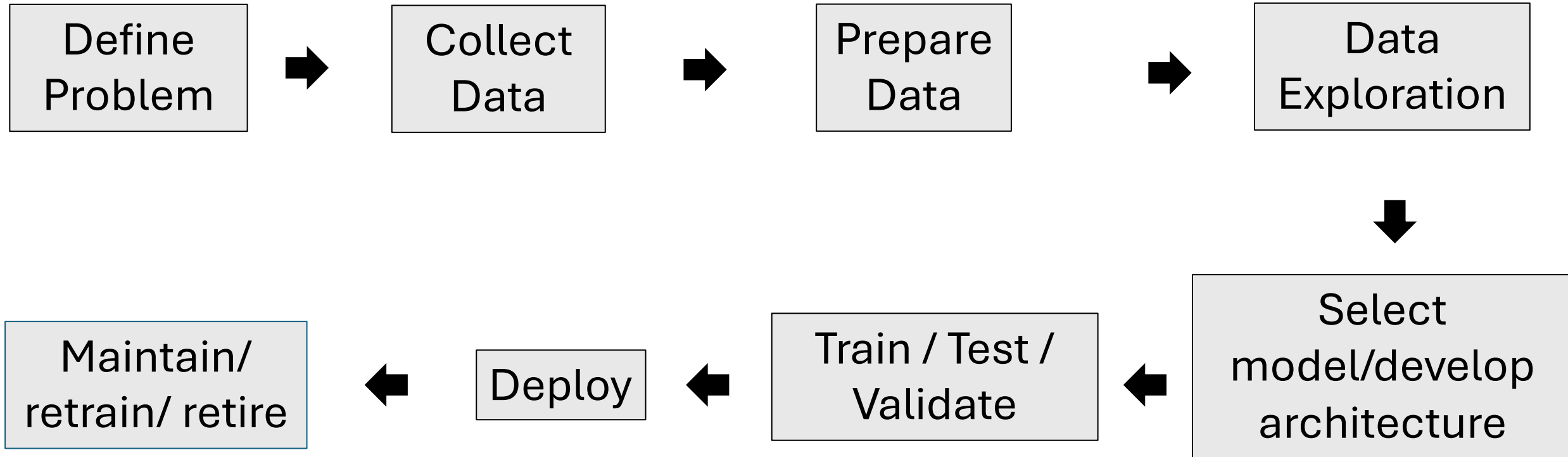


Pineapple

Inference: Use trained AI model to make predictions

ML Activity

Key steps



Why visualization?

Calculate the average of X

$X = [18, 34, 59, 32, 67]$



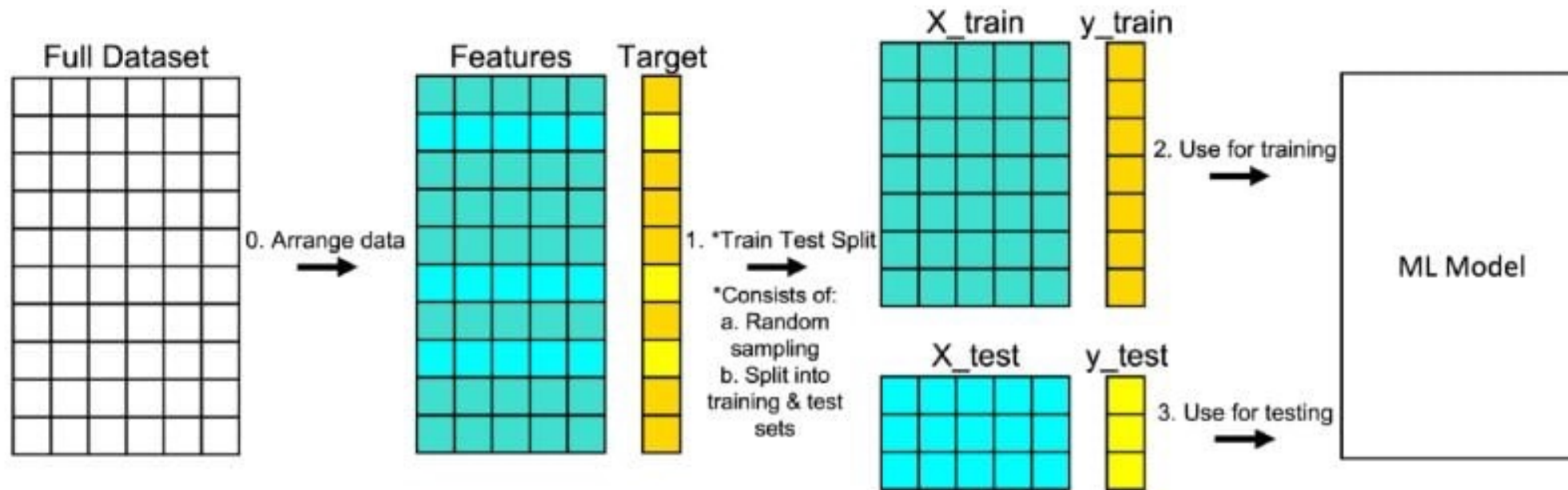
Why

Brain processes visual info way faster
than text, numbers or math

Cognitive load

Data visualization helps us digest
data

Training strategy with train/test split



70/30, 80/20 are commonly used strategies in ML

AI Platforms

LAYER	POPULAR LIBRARIES / MODELS	EXAMPLE USE CASES	KEY TAKEAWAY	DATA PRIVACY
1  MACHINE LEARNING (ML)	 - Classical ML algorithms - Structured data - Prediction & classification	 Sales forecasting  Customer segmentation  Churn prediction  Fraud detection	 Best for structured data & traditional ML problems. Focus: prediction & classification	 Highest data security Usually developed and run locally / on-premise. Sensitive data can stay inside the organization.
2  DEEP LEARNING (DL)	 - Neural networks - Unstructured data - High flexibility	 Image classification  Speech recognition  Video analysis  Medical image analysis	 Best for unstructured data (images, text, audio, video). Focus: representation learning & complex patterns	 Highest data security Often built and deployed internally. Strong control over training data, models, and access.
3  GENERATIVE AI (GenAI)	LEADING MODELS & PROVIDERS  Anthropic Claude  OpenAI ChatGPT  Microsoft Copilot  Gemini  Llama  Mistral AI  Cohere  Stability AI	 Content generation  Code generation  Chatbots & assistants  Image generation  Summarization & translation Audio & video generation	 Best for creating new content & human-like interactions. Focus: generation & augmentation	 More cloud-based risk Often accessed via external cloud services. Proper governance, access controls, and data policies are needed.

AI models in the Microsoft ecosystem

What you can access depends on the layer you use

1) Microsoft 365 Copilot

Word • Excel • PowerPoint • Outlook • Teams



GPT

Main/default model family

Claude

Available in supported Copilot experiences

Specialized
Microsoft-hosted models

Used behind specific features



Typical user experience: narrower model choice

2) Copilot Studio

Build business agents



GPT

Claude

Mistral

Grok

Choose the primary model for your agent



Broader than Microsoft 365 Copilot

3) Microsoft Foundry / Azure AI

Developer + enterprise platform



GPT

Claude

Phi

Llama

Mistral

DeepSeek

Grok

Cohere

FLUX

+ many more

Widest model catalogue



Best for custom apps, APIs, and advanced deployment